

Phase I/II Operating Characteristics

TITE versus IPDE-first non-TITE: one-stage, two-stage highest utility, and two-stage top-2 randomized

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 1						
DLT rate	0.3	0.35	0.4	0.45	0.5	
Efficacy rate	0.05	0.06	0.07	0.08	0.09	
OBD selection (%)						
TITE one-stage	19.1	8.4	1.9	0.3	0.1	70.2
IPDE-first non-TITE one-stage	18	8.2	2.5	0.4	0	70.9
TITE 2-stage highest utility	19.4	7.7	1.2	0.3	0	71.4
IPDE-first non-TITE 2-stage highest utility	18.1	8.3	1	0.4	0	72.2
TITE 2-stage top-2 randomized	18.7	8.7	3.9	0.7	0.2	67.8
IPDE-first non-TITE 2-stage top-2 randomized	17.8	7.8	2	0.8	0.1	71.5
Mean administrations by dose						
TITE one-stage	11.6	4.7	2	0.9	0.3	
IPDE-first non-TITE one-stage	10.7	4.8	2.6	1.1	0.3	
TITE 2-stage highest utility	11.5	4.6	2.1	0.9	0.3	
IPDE-first non-TITE 2-stage highest utility	10.7	4.7	2.7	1.1	0.3	
TITE 2-stage top-2 randomized	11.1	4.5	2	0.8	0.3	
IPDE-first non-TITE 2-stage top-2 randomized	10.6	4.8	2.6	1.1	0.3	
Mean trial duration (days)			Mean duration (days)			
TITE one-stage			711.6			
IPDE-first non-TITE one-stage			843.3			
TITE 2-stage highest utility			710.3			
IPDE-first non-TITE 2-stage highest utility			847.3			
TITE 2-stage top-2 randomized			690.1			
IPDE-first non-TITE 2-stage top-2 randomized			837.5			

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	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 16						
DLT rate	0.02	0.06	0.09	0.14	0.26	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
OBD selection (%)						
TITE one-stage	30.1	22.1	20.7	15	12	0.1
IPDE-first non-TITE one-stage	30	23.6	19.2	15.3	11.9	0
TITE 2-stage highest utility	32.6	23	20.1	14.2	10	0.1
IPDE-first non-TITE 2-stage highest utility	31.1	23.4	17.9	15.9	11.7	0
TITE 2-stage top-2 randomized	28.1	24.9	19.1	17.5	10.3	0.1
IPDE-first non-TITE 2-stage top-2 randomized	30.1	22.3	19.5	17.1	11	0
Mean administrations by dose						
TITE one-stage	7.1	6.7	5.9	5.7	4.6	
IPDE-first non-TITE one-stage	7.1	6.6	6	5.4	4.9	
TITE 2-stage highest utility	7.4	6.9	5.9	5.5	4.4	
IPDE-first non-TITE 2-stage highest utility	7.3	6.4	6.1	5.4	4.7	
TITE 2-stage top-2 randomized	6.1	6.1	5.6	5.7	5.3	
IPDE-first non-TITE 2-stage top-2 randomized	6.3	6	5.9	6.1	5.8	
Mean trial duration (days)			Mean duration (days)			
TITE one-stage			1082.9			
IPDE-first non-TITE one-stage			1248.7			
TITE 2-stage highest utility			1079.7			
IPDE-first non-TITE 2-stage highest utility			1248.4			
TITE 2-stage top-2 randomized			1070.7			
IPDE-first non-TITE 2-stage top-2 randomized			1247			

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	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 20						
DLT rate	0.07	0.14	0.31	0.41	0.52	
Efficacy rate	0.05	0.25	0.25	0.25	0.25	
OBD selection (%)						
TITE one-stage	11.2	45.4	27.3	11.9	1.6	2.6
IPDE-first non-TITE one-stage	13.5	44.3	26	11.8	2.6	1.8
TITE 2-stage highest utility	12.3	49.8	25.6	8	1.4	2.9
IPDE-first non-TITE 2-stage highest utility	11.2	48.2	28.2	8.8	1.6	2
TITE 2-stage top-2 randomized	8.4	49.3	25.4	12.7	1.6	2.6
IPDE-first non-TITE 2-stage top-2 randomized	9.1	49.3	25.1	11.5	2.4	2.6
Mean administrations by dose						
TITE one-stage	5.2	11	7.7	4.2	1.4	
IPDE-first non-TITE one-stage	5.2	10.7	7.8	4.3	1.7	
TITE 2-stage highest utility	5.3	11.1	8.1	3.7	1.4	
IPDE-first non-TITE 2-stage highest utility	5.3	10.9	8.2	3.8	1.5	
TITE 2-stage top-2 randomized	5.7	8.6	7.2	4.2	1.7	
IPDE-first non-TITE 2-stage top-2 randomized	5.7	9.4	8	4.7	1.9	
Mean trial duration (days)			Mean duration (days)			
TITE one-stage			1100.7			
IPDE-first non-TITE one-stage			1274			
TITE 2-stage highest utility			1102.4			
IPDE-first non-TITE 2-stage highest utility			1275.9			
TITE 2-stage top-2 randomized			1088.9			
IPDE-first non-TITE 2-stage top-2 randomized			1280			

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	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 24						
DLT rate	0.01	0.02	0.05	0.1	0.25	
Efficacy rate	0.05	0.08	0.2	0.5	0.5	
OBD selection (%)						
TITE one-stage	0.9	1.1	4.5	60	33.5	0
IPDE-first non-TITE one-stage	1.6	0.8	5.4	59.5	32.7	0
TITE 2-stage highest utility	1	1.5	4.8	56.3	36.4	0
IPDE-first non-TITE 2-stage highest utility	1.7	0.9	6.1	55	36.3	0
TITE 2-stage top-2 randomized	0.8	1.1	4.1	54.3	39.7	0
IPDE-first non-TITE 2-stage top-2 randomized	1	0.8	5.3	53.9	39	0
Mean administrations by dose						
TITE one-stage	3.4	3.5	4.5	12.2	6.5	
IPDE-first non-TITE one-stage	3.5	3.4	4.5	12.2	6.4	
TITE 2-stage highest utility	3.4	3.5	4.5	10.6	8	
IPDE-first non-TITE 2-stage highest utility	3.5	3.4	4.7	10.6	7.9	
TITE 2-stage top-2 randomized	3.5	3.7	4.9	9.1	8.4	
IPDE-first non-TITE 2-stage top-2 randomized	3.6	3.7	5.1	9.2	8.4	
Mean trial duration (days)			Mean duration (days)			
TITE one-stage			1202.6			
IPDE-first non-TITE one-stage			1321.1			
TITE 2-stage highest utility			1200.9			
IPDE-first non-TITE 2-stage highest utility			1328.9			
TITE 2-stage top-2 randomized			1183			
IPDE-first non-TITE 2-stage top-2 randomized			1321.3			

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	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 27						
DLT rate	0.11	0.17	0.3	0.5	0.62	
Efficacy rate	0.1	0.2	0.4	0.42	0.45	
OBD selection (%)						
TITE one-stage	13.1	25.4	45	10.9	0.6	5
IPDE-first non-TITE one-stage	15.6	21.2	47.9	10.6	0.5	4.2
TITE 2-stage highest utility	15.8	28.1	43	6.8	0.4	5.9
IPDE-first non-TITE 2-stage highest utility	17.1	26.9	43.5	6.5	0.7	5.3
TITE 2-stage top-2 randomized	11.8	25.9	42.7	13.6	1.1	4.9
IPDE-first non-TITE 2-stage top-2 randomized	11.3	23.9	46.5	12.7	1.3	4.3
Mean administrations by dose						
TITE one-stage	7.1	8.2	9.4	3.8	0.6	
IPDE-first non-TITE one-stage	7.1	7.8	10.2	3.5	0.7	
TITE 2-stage highest utility	7.7	8.2	9	3.3	0.9	
IPDE-first non-TITE 2-stage highest utility	7.7	8.2	9.4	3.2	0.9	
TITE 2-stage top-2 randomized	7.1	7.6	7.4	3.6	0.9	
IPDE-first non-TITE 2-stage top-2 randomized	7.5	8.2	8.5	4	1.1	
Mean trial duration (days)			Mean duration (days)			
TITE one-stage			1124.4			
IPDE-first non-TITE one-stage			1304.9			
TITE 2-stage highest utility			1117.6			
IPDE-first non-TITE 2-stage highest utility			1307.7			
TITE 2-stage top-2 randomized			1083.5			
IPDE-first non-TITE 2-stage top-2 randomized			1308.9			

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	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 38						
DLT rate	0.05	0.1	0.12	0.15	0.3	
Efficacy rate	0.05	0.08	0.12	0.15	0.5	
OBD selection (%)						
TITE one-stage	14.5	9.2	7.5	8.3	59.2	1.3
IPDE-first non-TITE one-stage	14.2	7.7	8	8.1	61.4	0.6
TITE 2-stage highest utility	16.5	10.2	9	9.2	53.6	1.5
IPDE-first non-TITE 2-stage highest utility	17.9	8.5	9.1	9.4	54.2	0.9
TITE 2-stage top-2 randomized	11.1	9.4	8.9	9.9	59.6	1.1
IPDE-first non-TITE 2-stage top-2 randomized	8.2	5.5	7.6	11.1	66.8	0.8
Mean administrations by dose						
TITE one-stage	4.7	4.9	5	5.4	9.9	
IPDE-first non-TITE one-stage	4.7	4.4	5.3	5.3	10.1	
TITE 2-stage highest utility	4.9	5	5.5	5.4	9.1	
IPDE-first non-TITE 2-stage highest utility	4.9	4.6	5.7	5.4	9.3	
TITE 2-stage top-2 randomized	4.6	4.8	5.1	6.1	7.1	
IPDE-first non-TITE 2-stage top-2 randomized	4.8	4.7	5.4	6.9	8	
Mean trial duration (days)			Mean duration (days)			
TITE one-stage			1133.8			
IPDE-first non-TITE one-stage			1287			
TITE 2-stage highest utility			1123			
IPDE-first non-TITE 2-stage highest utility			1277.4			
TITE 2-stage top-2 randomized			1067.3			
IPDE-first non-TITE 2-stage top-2 randomized			1263.3			